

A  
**PROJECT REPORT**  
ON

**PRAVA**  
**College Academic Management System**

Developed for  
**Women's College of Home Science and BCA, Loni**

Submitted to  
**Savitribai Phule Pune University, Pune**

Submitted by  
**Rutuja Ashok Khobare | Roll No. 30**

Under the Guidance of  
**Surekha Kale**

For the Academic Year  
**2026–2027**

In partial fulfilment of the requirements for the degree of  
**Bachelor of Computer Applications**  
**DEPARTMENT OF BCA**



Loknete Dr. Balasaheb Vikhe Patil (Padma Bhushan Awardee)

Pravara Rural Education Society's

**Women's College of Home Science and BCA, Loni**

PKVM Campus, Babhaleshwar Road, Loni-413713

Dist-Ahmednagar, Maharashtra, India

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**Academic Year 2026–2027**

# PROJECT SUBMISSION

Submitted by

**Rutuja Ashok Khobare | Roll No. 30**

Student of TY BCA – Third Year

**Bachelor of Computer Applications**

I submit the group project

**PRAVA**

**College Academic Management System**

I, a student of TY BCA, submit the group project entitled “PRAVA – College Academic Management System” for the academic year 2026–2027.

The project presents the design and implementation of an academic management portal for students, teaching staff and college administration. This report documents the application modules, database design, user interfaces and verification of the implemented system.

I have prepared this project as a group member under the guidance of Surekha Kale as part of the Bachelor of Computer Applications project work at Savitribai Phule Pune University, Pune.

Place: \_\_\_\_\_

Date: \_\_\_\_\_

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**Rutuja Ashok Khobare**

Student signature | Roll No. 30

**Loknete Dr. Balasaheb Vikhe Patil**  
**(Padma Bhushan Awardee)**

Pravara Rural Education Society's

**Women's College of Home Science and BCA, Loni**

## **CERTIFICATE**

This is to certify that

**Rutuja Ashok Khobare | Roll No. 30**

a student of TY BCA – Third Year, has prepared and submitted  
the group project

**PRAVA**

**College Academic Management System**

under the guidance of Surekha Kale, in partial fulfilment  
of the requirements for the degree of

**Bachelor of Computer Applications**

Savitribai Phule Pune University, Pune

**Academic Year 2026–2027**

\_\_\_\_\_  
**Dr. Sanjay Vani**

Head of Department

\_\_\_\_\_  
**External Examiner**

Name and signature

\_\_\_\_\_  
**Surekha Kale**

Project Guide

\_\_\_\_\_  
**Dr. Anushree Rajendra Khaire**

Principal

**College Seal**

Place: \_\_\_\_\_

Date: \_\_\_\_\_

## **STUDENT DECLARATION**

I, Rutuja Ashok Khobare (Roll No. 30), a student of TY BCA at Women's College of Home Science and BCA, Loni, declare that this report presents my work on the group project “PRAVA – College Academic Management System” for the academic year 2026–2027.

The report describes the design and implementation of the college academic portal, including student, staff and office workspaces, database records, academic workflows and access controls. It is submitted under the guidance of Surekha Kale as part of my Bachelor of Computer Applications project work.

College information, university curriculum documents, open-source software and technical references are acknowledged in the bibliography. Generated local records illustrate academic workflows; they do not certify college results. Submitted student profiles are retained separately from these presentation records.

Class: TY BCA – Third Year

Project Type: Group Project

Academic Year: 2026–2027

Place: \_\_\_\_\_

Date: \_\_\_\_\_

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**Rutuja Ashok Khobare**

Student signature | Roll No. 30

## **ACKNOWLEDGEMENT**

I thank my project guide, Surekha Kale, for guidance in the preparation of my TY BCA project report, PRAVA – College Academic Management System.

I express my gratitude to Dr. Sanjay Vani, Head of Department, and Dr. Anushree Rajendra Khaire, Principal, for the academic environment in which this project is presented. I also thank the teaching staff of Women's College of Home Science and BCA, Loni for supporting my learning.

The project gave me an opportunity to connect web development with a practical college workflow. Preparing separate student, staff and office workspaces helped me understand how interface design, validation, database relationships and access permissions work together.

I acknowledge the college website, university syllabus documents, open-source software and official technical documentation used during the development and documentation of the project. These resources helped me describe the academic context and implement the application.

I thank my family, friends and everyone who assisted me with project review and demonstration preparation.

With sincere thanks,

**Rutuja Ashok Khobare | Roll No. 30**

TY BCA – Academic Year 2026–2027

## Technology Profile

PRAVA is a browser-based academic portal. A Python and Flask application manages accounts, validates academic operations and stores structured records in a relational database. The local presentation version uses SQLite and serves the website on the same computer.

| Layer          | Technology                        | Purpose                                              |
|----------------|-----------------------------------|------------------------------------------------------|
| Presentation   | HTML, CSS, JavaScript             | Responsive public pages and role-specific workspaces |
| Templates      | Jinja                             | Server-rendered pages with shared navigation         |
| UI library     | Bootstrap 5.3.3 and Icons 1.11.3  | Locally bundled layout and interface assets          |
| Application    | Python 3.11–3.13; Flask 3.0.3     | Routes, services and application configuration       |
| Persistence    | Flask-SQLAlchemy 3.1.1; SQLite    | Models, relationships and local records              |
| Authentication | Flask-Login 0.6.3; Werkzeug 3.0.3 | Sessions and password-hash verification              |
| Forms          | Flask-WTF 1.2.1 and WTForms 3.1.2 | Input validation and CSRF protection                 |
| Testing        | Python unittest                   | Isolated automated functional and security checks    |
| Delivery       | GitHub and Windows launcher       | Versioned source and repeatable local setup          |

The repository also contains PostgreSQL and optional external authentication configuration for future hosting. This report evaluates the working local-authentication installation. Package versions above describe the project requirements and bundled assets.

## **Abstract**

PRAVA is a College Academic Management System for Women's College of Home Science and BCA, Loni. It connects public college information with student, staff and administrative workspaces. The project supports BCA and the Home Science specialisations in Food Science and Nutrition, and Textile Science and Designing, using a versioned 2024 NEP curriculum catalogue.

The website opens on Home. Its header contains Home, Academics, Contact and a Login dropdown. Student login continues through BCA or Home Science selection, while Staff and Office / Administration open their own forms. The authenticated account determines the displayed name, permissions and dashboard. Night mode is the default, with a Day mode option.

Students can submit a website sign-up request for Office approval. Separate Google Forms collect student and staff profile information into private response spreadsheets. Those responses require verification and conversion to the portal CSV templates; submitting a Google Form does not automatically create a PRAVA account.

Python, Flask, Jinja and SQLAlchemy implement the application, while a local SQLite database preserves users and academic records. The 22 model tables support profiles, teaching allocations, attendance, marks, materials, assignments, notices, curriculum sources, activities and review workflows. Password hashing, CSRF protection, role checks and record ownership restrict access.

The repository baseline is 4abb237 with local typography improvements. All 63 automated tests passed on 5 October 2026. This report documents the implementation, database, academic workspaces and setup. Final-year syllabus verification, official NEP grade calculations and account notifications remain future enhancements.

Keywords: academic management, Flask, SQLite, BCA, Home Science, NEP curriculum, role-based access.

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# **1 Introduction Objectives and Scope**

College academic records connect identities, programme enrolment, teaching allocations, attendance, assessments and learning resources. Students need access to their own progress; teachers need the records of assigned classes; and the Office needs consistent institutional data. PRAVA brings these responsibilities into one browser-based application.

## **Problem and proposed system**

Keeping the same academic information in separate registers, files and messages can make retrieval slow and introduce duplicate entries. PRAVA addresses this design problem through shared relational records and server-side access rules. Public college content, pending registrations and verified academic profiles are maintained separately.

## **Objectives**

- Provide clear public navigation and role-specific login with the correct account identity.
- Verify student registration and roster imports before activating academic accounts.
- Link student and staff work to the correct programme, curriculum, semester and subject.
- Manage attendance, marks, materials, assignments, notices, activities and reports.
- Provide source-linked syllabus information and a repeatable local demonstration setup.

## **Project scope**

The catalogue covers BCA, B.Sc. Food Science and Nutrition, and B.Sc. Textile Science and Designing. FY, SY and TY navigation represents the three-year journey; detail is displayed according to the available source review. Office and Principal access use distinct administrative permissions. The local project demonstrates academic operations; fees, payroll, library circulation and university result certification are outside its current modules.

## 2 Requirements Analysis and Feasibility

The requirements come from the requested college workflows, the supplied report format, college and curriculum references, and source-code inspection. Google Forms support profile collection. Submitted records require Office review; the project does not claim measured adoption benefits or institutional-scale operation.

| Requirement        | Local demonstration setup                                                                                   |
|--------------------|-------------------------------------------------------------------------------------------------------------|
| Hardware           | 64-bit laptop or desktop; 4 GB RAM and 1 GB free space are planning recommendations, not measured minima.   |
| Software           | Python 3.11–3.13, a modern browser and the pinned requirements.txt dependencies.                            |
| Database and files | SQLite in instance/prava.sqlite3; learning files and profile uploads stored separately.                     |
| Connectivity       | Internet for first setup and external sources. Bundled local pages render after setup without internet.     |
| Quality            | Readable Day/Night themes, validation feedback, persistent records, controlled access and auditable review. |

### Feasibility

Technical feasibility follows from the conventional browser, Flask and relational database architecture. SQLite supports local presentation without a separate database server. Operational feasibility depends on the Office verifying identities and allocating subjects before Staff enters records. Separate dashboards make these responsibilities visible.

An existing laptop and open-source dependencies support an economical local demonstration. Printing, binding and any later hosting remain separate costs. Data feasibility depends on official student identifiers, staff assignments and applicable batch documents. Missing academic results remain pending, while unresolved curriculum items retain review status.

### 3 System Architecture and Modules

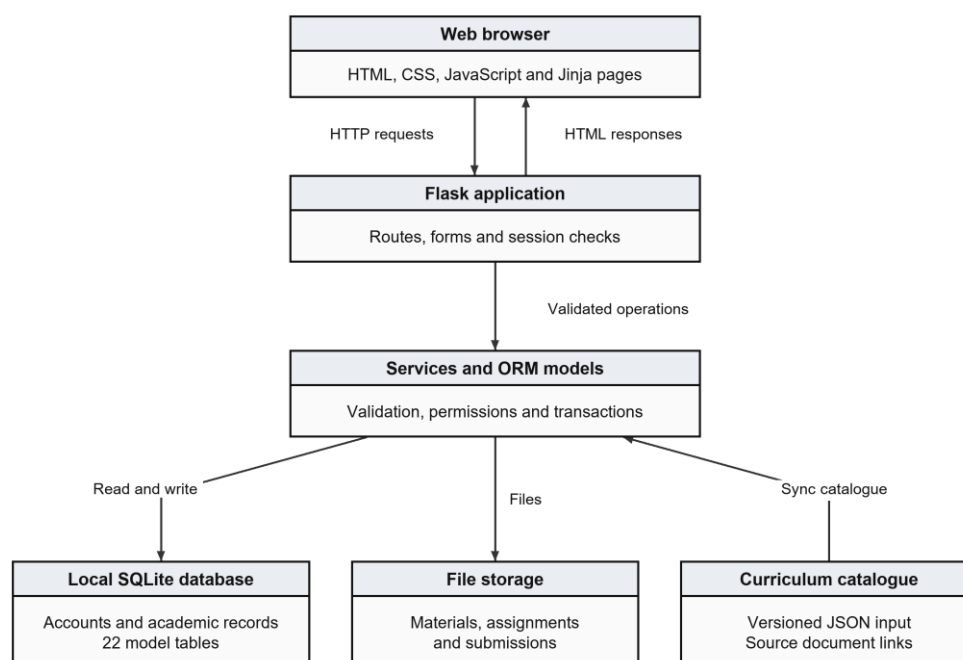


Figure 1 Browser application services and local storage

Flask routes receive requests, validate forms and check sessions and roles. Services perform academic operations through SQLAlchemy. Jinja renders HTML responses; uploaded files remain separate from their metadata. The curriculum JSON synchronises reference information into the database.

| Module            | Principal responsibility                                                                   |
|-------------------|--------------------------------------------------------------------------------------------|
| Public and access | Home, college information, academics, contact, role login and student sign-up.             |
| Student and Staff | Programme-based progress views; assigned-subject attendance, marks and learning resources. |
| Administration    | Profiles, approvals, curriculum review, imports, notices, access scopes and exports.       |
| Shared services   | Database transactions, validation, activity records, upload checks and persistent setup.   |

## 4 Login Registration and Permissions

| Entry point                  | Destination and rule                                                                              |
|------------------------------|---------------------------------------------------------------------------------------------------|
| Home → Login dropdown        | Student, Staff and Office / Administration appear immediately below Login.                        |
| Student → BCA / Home Science | Dedicated form validates credentials, active status, role and programme.                          |
| Staff                        | Staff credentials open the faculty workspace and assigned-subject tools.                          |
| Office / Administration      | Administrative workspace follows Office, Principal or administrator scope.                        |
| Back Switch and Logout       | Back uses defined internal pages. Switching requires new credentials. Logout uses protected POST. |

### Student sign-up and approval

Website sign-up captures identity, programme, semester, admission year and a password that is stored only as a hash. A valid request becomes Pending in student\_registrations. Office approval rechecks duplicate identities and creates the user and student profile in one transaction. Rejection preserves the review decision without activating an account.

### Permission boundaries

A student reads her own academic records. Staff works with explicitly assigned subjects and eligible students. Office manages authorised records and approvals. Principal scope is read-only for permitted oversight. Privileged account creation requires administrator scope. A failed account switch preserves the earlier authenticated identity; choosing a portal alone does not change the account.

Public student sign-up currently uses local password authentication. Optional external authentication requires additional provisioning work. Registration approval emails are not automated.

## Use Case Diagram

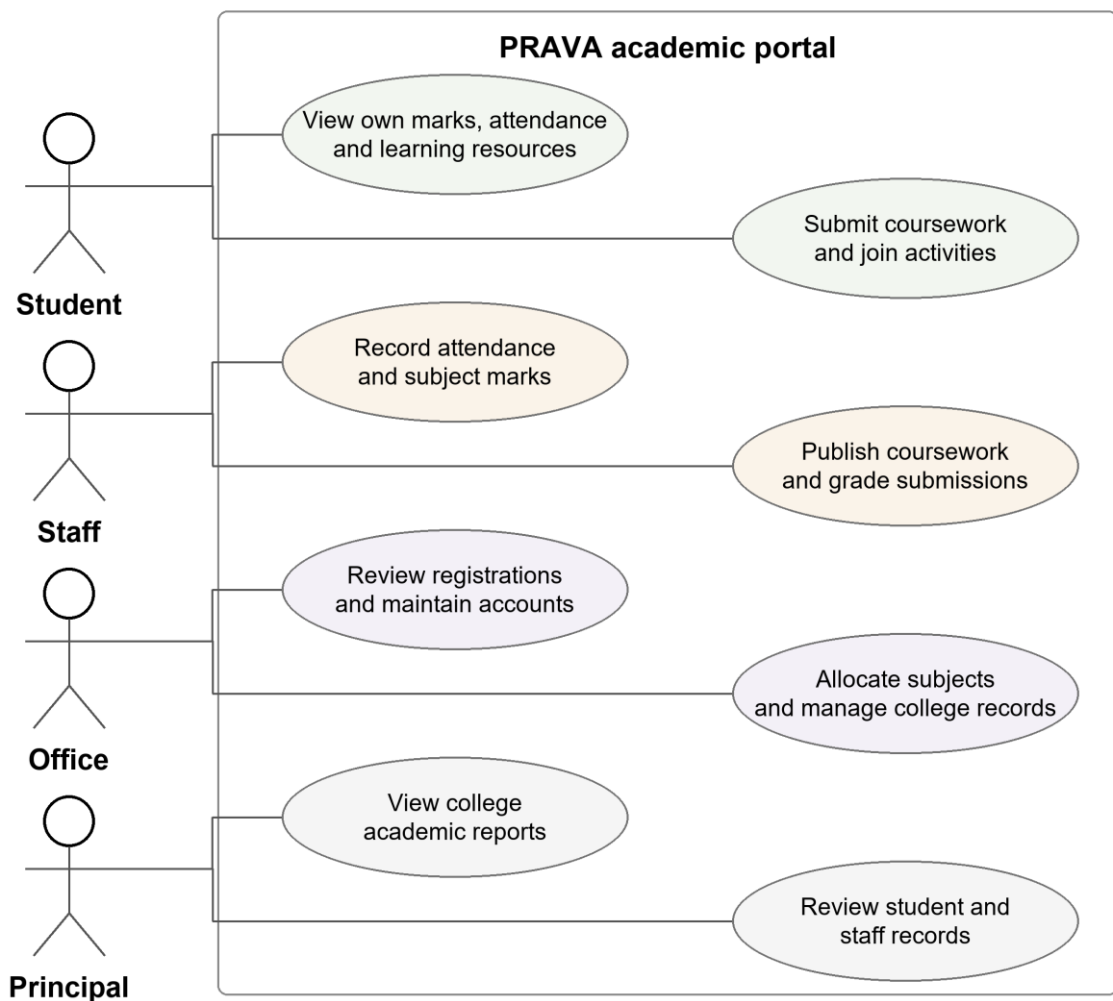


Figure 2 Primary actors and their academic operations

Student access is restricted to her own records. Staff operations depend on subject allocation. Office handles record maintenance and registration review; the Principal account provides read-only oversight.

Protected operations require an active authenticated account. A prospective student may submit a public sign-up request; Office approval is required before that account can access the student workspace.

## 5 Database Design and ER Diagrams

The implementation contains 22 model tables and 227 columns. PK denotes a primary key, FK a foreign key and UQ a uniqueness constraint. A 1-to-many relationship is implemented by a child foreign key; 1-to-0..1 describes an optional unique profile for a user.

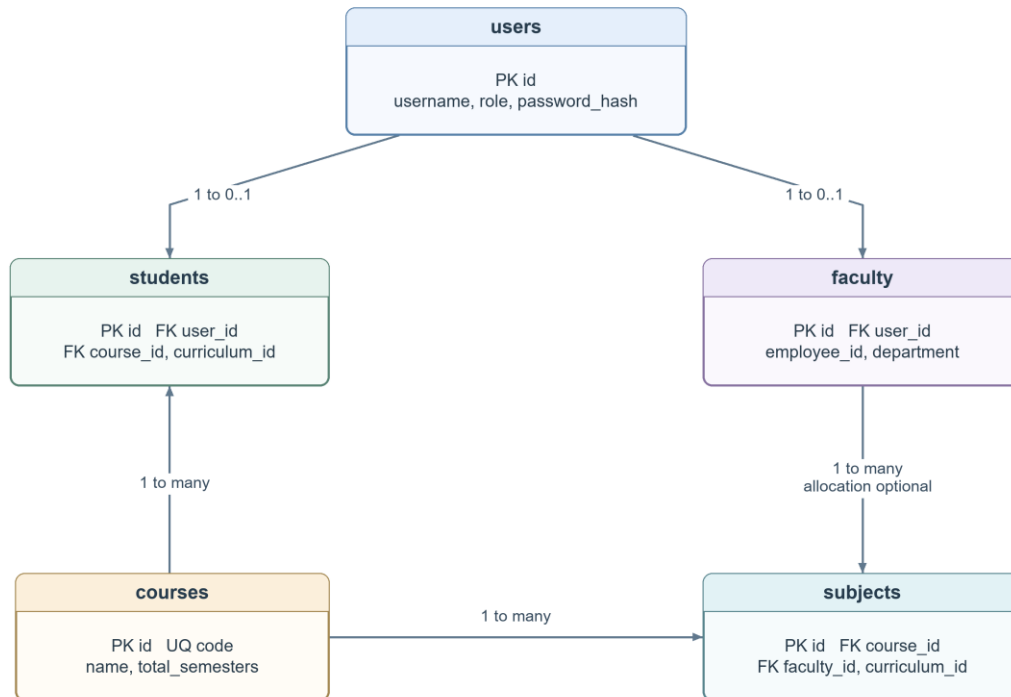


Figure 3 Accounts profiles courses and teaching subjects

users holds authentication identity; students and faculty hold role-specific profiles. Students belong to courses and curricula. subjects describes actual teaching allocations, including programme, semester and faculty. The separate curriculum catalogue provides source-reviewed reference entries.

The data dictionary follows SQLAlchemy declarations. Application validation and SQLite enforcement can differ. Password fields contain hashes; JSON text fields hold structured data. Foreign-key relationships are shown only where declared.

## Academic and Learning Relationships

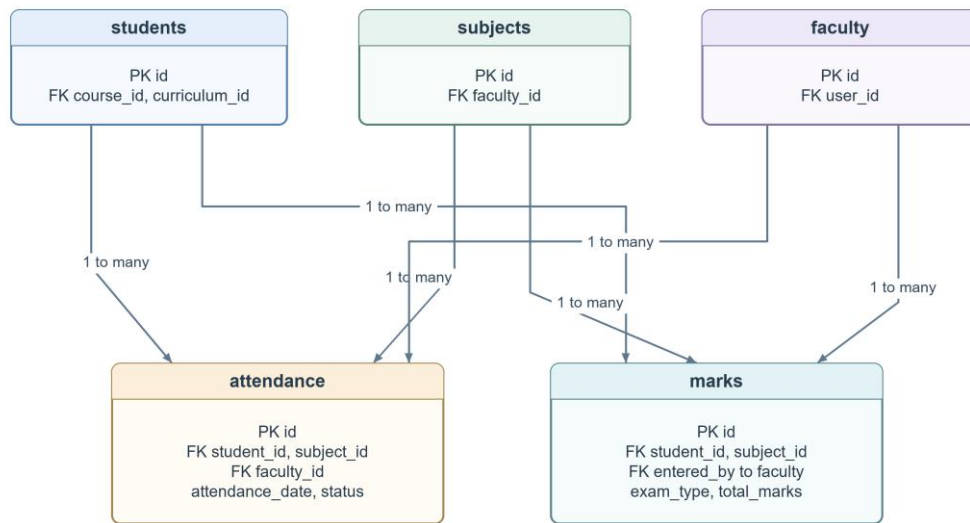


Figure 4 Student subject and faculty references in assessment

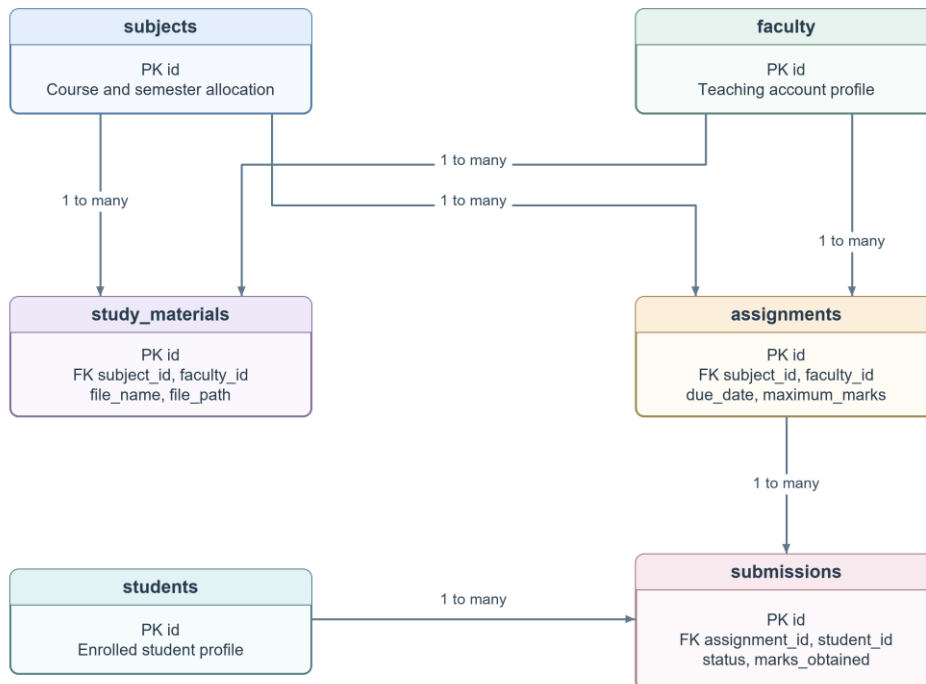


Figure 5 Learning materials assignments and submissions

Attendance is unique per student, subject and date; marks per student, subject and exam type. An assignment has at most one tracked submission per student. All academic operations also check the authorised subject relationship.

## Curriculum and Participation Relationships

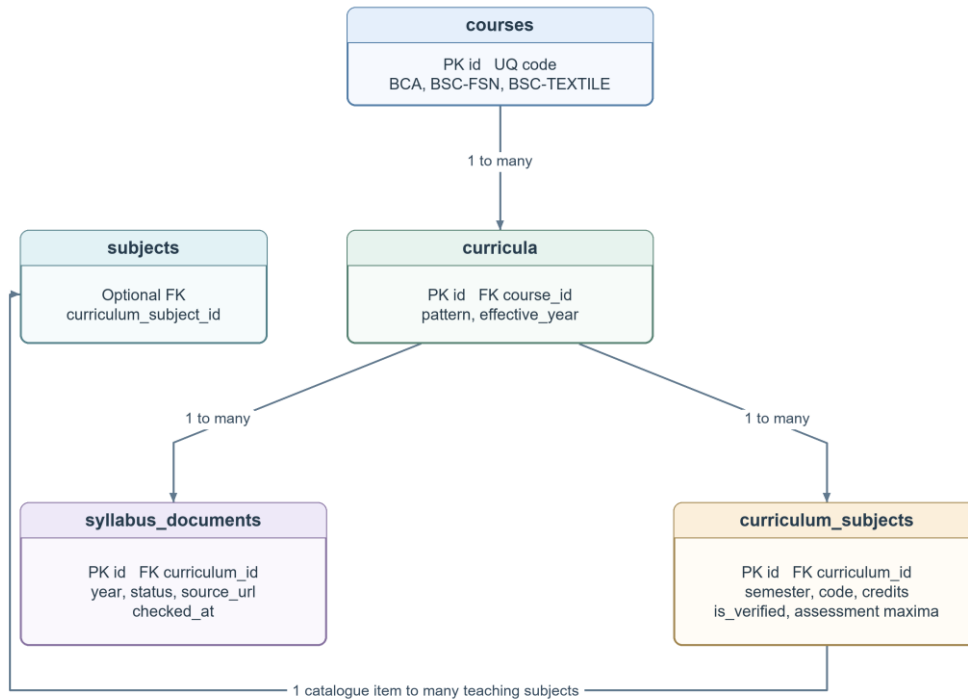


Figure 6 Versioned curriculum documents and catalogue subjects

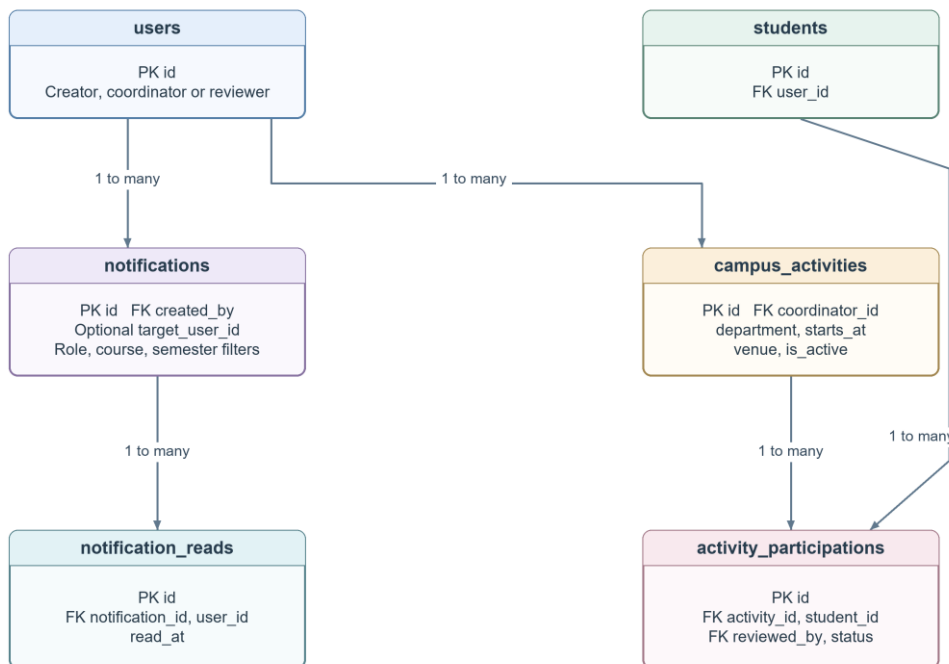


Figure 7 Notifications read receipts and activity participation

Source documents and catalogue subjects belong to a curriculum pattern. Activity participation links a student to an activity and its review. Contact inquiries and student-test snapshots are independent records; their models do not declare a student-account foreign key.

## 6 Data Flow Diagrams

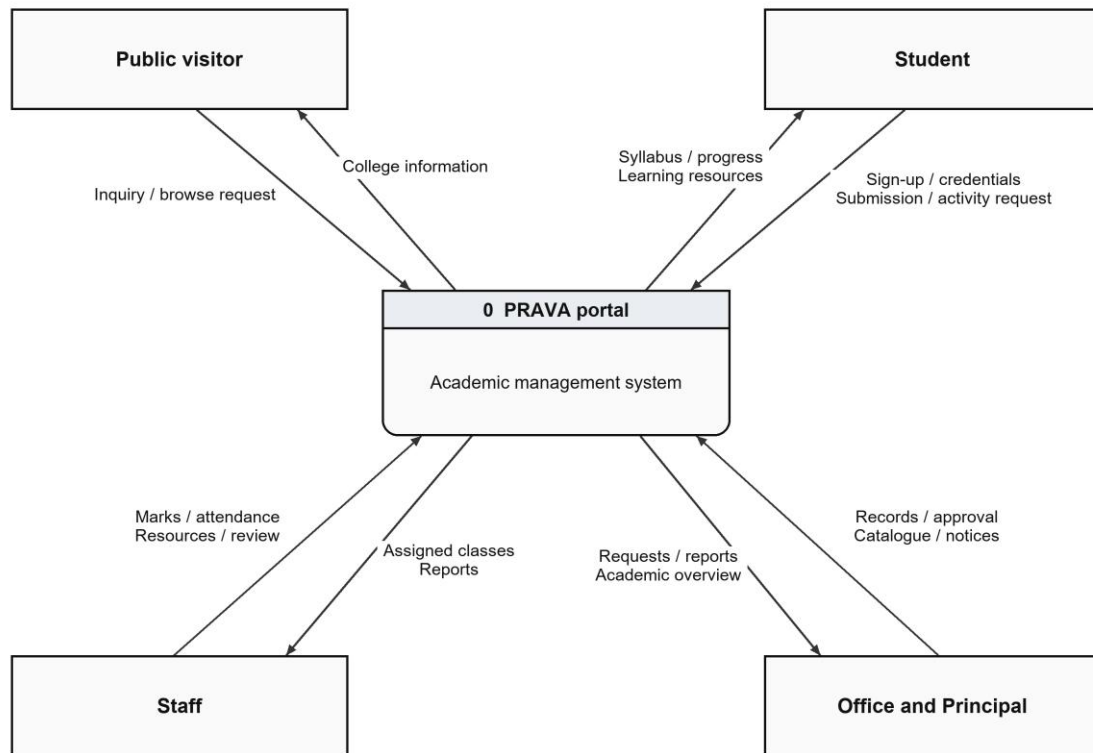


Figure 8 Context DFD for the PRAVA academic portal

The context diagram shows the system boundary. Visitors request public information and submit inquiries. Students supply registration and learning inputs and receive their academic views. Staff maintain assigned academic records. Office manages approvals and institutional records; Principal access provides permitted oversight.

Arrows identify information exchanged with the system. The database is internal to PRAVA and is expanded into logical stores at the next level. Google Forms collection is a separate preparatory intake process, followed by verified CSV import.

## Level One Data Flow

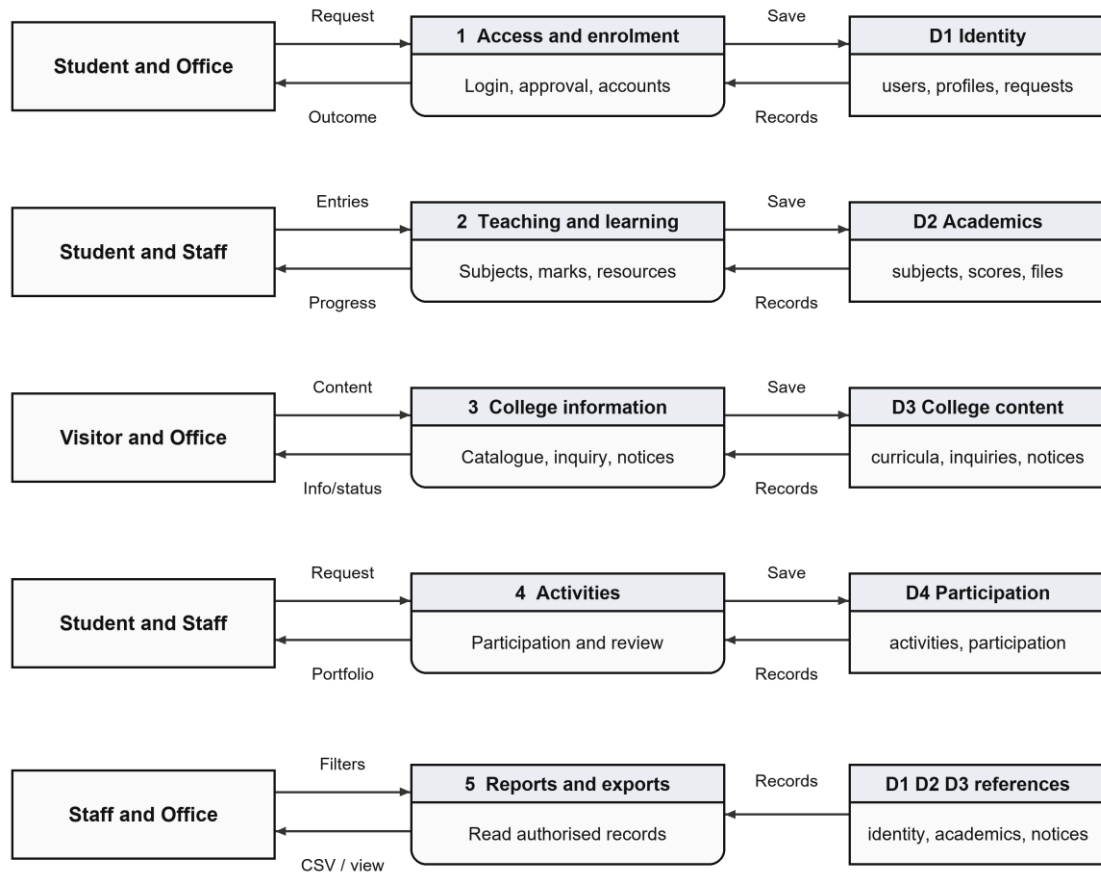


Figure 9 Level one decomposition and logical data stores

The five processes cover access and enrolment, teaching and learning, college information, activities, and reporting. D1–D4 group related tables in the same relational database. Reporting reads existing records within the current account scope and produces views or CSV files.

A report does not create a second independent academic total. The data stores are logical groups; the application maintains a shared schema with database keys and service-level validation.

## Level Two Registration Review

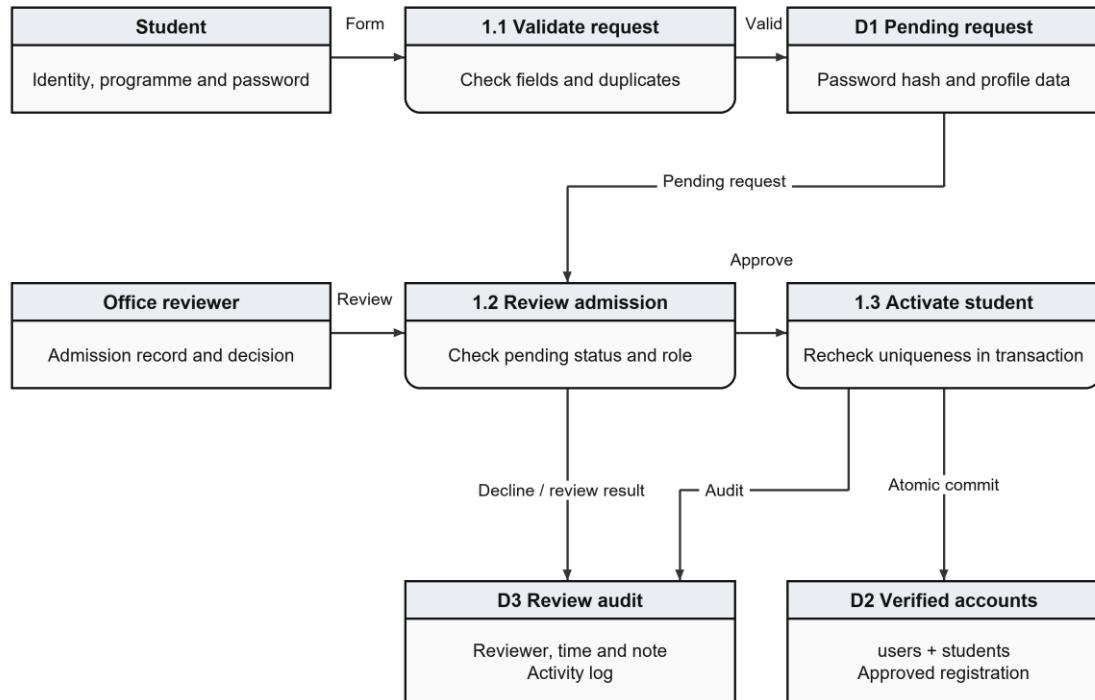


Figure 10 Student sign-up validation Office review and activation

A valid website sign-up creates a pending request after format and duplicate checks. An authorised Office user reviews the request. Approval repeats the uniqueness checks and creates a login identity and student profile atomically. A conflict rolls the operation back and returns review feedback.

The review records its decision, reviewer and time. Rejected requests remain reviewed. Approval permits the student to sign in with the chosen credentials; it does not automatically send an email or create subject allocations.

## 7 Data Dictionary

The schema contains 22 tables and 227 declared columns. This five-page summary lists every table and selected fields needed to explain the principal workflows. Repeated audit timestamps and secondary profile fields remain defined in the source models.

| Table                   | Purpose                                        |
|-------------------------|------------------------------------------------|
| users                   | Login identity, role, scope and active status. |
| students                | Student enrolment and programme profile.       |
| faculty                 | Staff identity and professional profile.       |
| courses                 | Degree programmes and semester count.          |
| subjects                | Teaching subjects and faculty allocation.      |
| attendance              | Date-wise student attendance.                  |
| marks                   | Subject assessments and component scores.      |
| assignments             | Faculty coursework and due dates.              |
| submissions             | Student work, marks and feedback.              |
| study_materials         | Uploaded learning-resource metadata.           |
| student_registrations   | Pending sign-ups and Office decisions.         |
| curricula               | Programme pattern and effective year.          |
| curriculum_subjects     | NEP catalogue subjects and assessment limits.  |
| syllabus_documents      | Year-wise source links and review status.      |
| campus_activities       | College activities and coordinators.           |
| activity_participations | Student activity registration and review.      |
| notifications           | Targeted announcements and expiry.             |
| notification_reads      | Per-user notice read receipts.                 |
| contact_inquiries       | Public contact messages and status.            |
| record_imports          | Staged CSV import payload and owner.           |
| activity_logs           | Recorded user actions and timestamps.          |
| student_test_responses  | Student test answers, score and feedback.      |

Notation: PK = primary key; FK = foreign key; UQ = unique. Each table has an id primary key. Field types and keys below follow the SQLAlchemy declarations; validation rules are described separately.

## Identity and Enrolment

| Table    | Field             | Type         | Key / meaning                            |
|----------|-------------------|--------------|------------------------------------------|
| users    | id                | INTEGER      | PK                                       |
|          | username          | VARCHAR(80)  | UQ                                       |
|          | full_name         | VARCHAR(120) | Account holder name                      |
|          | email             | VARCHAR(120) | UQ                                       |
|          | password_hash     | VARCHAR(255) | One-way password hash                    |
|          | role              | VARCHAR(20)  | student / faculty / admin                |
|          | admin_scope       | VARCHAR(30)  | Administrator / office / principal scope |
|          | is_active         | BOOLEAN      | Controls account availability            |
| students | id                | INTEGER      | PK                                       |
|          | user_id           | INTEGER      | UQ; FK users.id                          |
|          | enrollment_number | VARCHAR(50)  | UQ                                       |
|          | course_id         | INTEGER      | FK courses.id                            |
|          | curriculum_id     | INTEGER      | FK curricula.id                          |
|          | semester          | INTEGER      | Current / assigned semester              |
|          | admission_year    | INTEGER      | Year of admission                        |
| faculty  | id                | INTEGER      | PK                                       |
|          | user_id           | INTEGER      | UQ; FK users.id                          |
|          | employee_id       | VARCHAR(50)  | UQ                                       |
|          | department        | VARCHAR(120) | Teaching department                      |
|          | qualification     | VARCHAR(120) | Recorded qualification                   |

users stores shared authentication identity. A unique user\_id connects each student or faculty profile to one account. Enrolment numbers and employee IDs identify the corresponding college records.

## Teaching and Assessment

| Table      | Field                 | Type         | Key / meaning               |
|------------|-----------------------|--------------|-----------------------------|
| courses    | id                    | INTEGER      | PK                          |
|            | code                  | VARCHAR(30)  | UQ                          |
|            | name                  | VARCHAR(120) | Display name                |
|            | total_semesters       | INTEGER      | Programme semester count    |
| subjects   | id                    | INTEGER      | PK                          |
|            | code                  | VARCHAR(30)  | Programme / subject code    |
|            | course_id             | INTEGER      | FK courses.id               |
|            | semester              | INTEGER      | Current / assigned semester |
|            | faculty_id            | INTEGER      | FK faculty.id               |
|            | curriculum_subject_id | INTEGER      | FK curriculum_subjects.id   |
| attendance | id                    | INTEGER      | PK                          |
|            | student_id            | INTEGER      | FK students.id              |
|            | subject_id            | INTEGER      | FK subjects.id              |
|            | attendance_date       | DATE         | Date of the class record    |
|            | status                | VARCHAR(20)  | Workflow / attendance state |
| marks      | id                    | INTEGER      | PK                          |
|            | student_id            | INTEGER      | FK students.id              |
|            | subject_id            | INTEGER      | FK subjects.id              |
|            | exam_type             | VARCHAR(50)  | Assessment label            |
|            | internal_marks        | INTEGER      | Internal component          |
|            | external_marks        | INTEGER      | External component          |
|            | total_marks           | INTEGER      | Internal + external total   |

Composite uniqueness: subjects (code, course\_id, semester); attendance (student\_id, subject\_id, attendance\_date); marks (student\_id, subject\_id, exam\_type). Assessment services check eligibility and configured component limits.

## Coursework and Registration

| Table                 | Field             | Type         | Key / meaning               |
|-----------------------|-------------------|--------------|-----------------------------|
| assignments           | id                | INTEGER      | PK                          |
|                       | subject_id        | INTEGER      | FK subjects.id              |
|                       | faculty_id        | INTEGER      | FK faculty.id               |
|                       | due_date          | DATETIME     | Submission deadline         |
|                       | maximum_marks     | INTEGER      | Maximum assignment score    |
| submissions           | id                | INTEGER      | PK                          |
|                       | assignment_id     | INTEGER      | FK assignments.id           |
|                       | student_id        | INTEGER      | FK students.id              |
|                       | marks_obtained    | INTEGER      | Awarded assignment score    |
|                       | faculty_feedback  | TEXT         | Review comments             |
|                       | status            | VARCHAR(30)  | Workflow / attendance state |
| study_materials       | id                | INTEGER      | PK                          |
|                       | subject_id        | INTEGER      | FK subjects.id              |
|                       | file_path         | VARCHAR(255) | Stored resource path        |
| student_registrations | id                | INTEGER      | PK                          |
|                       | enrollment_number | VARCHAR(50)  | UQ                          |
|                       | curriculum_id     | INTEGER      | FK curricula.id             |
|                       | status            | VARCHAR(20)  | Workflow / attendance state |
|                       | reviewed_by       | INTEGER      | FK users.id                 |
|                       | user_id           | INTEGER      | UQ; FK users.id             |

One submission is tracked per assignment and student. Registration review activates an approved account and links user\_id; approval rechecks uniqueness. Uploaded-file paths identify controlled local storage, not a second academic database.

## Curriculum and Activities

| Table                   | Field          | Type          | Key / meaning               |
|-------------------------|----------------|---------------|-----------------------------|
| curricula               | id             | INTEGER       | PK                          |
|                         | course_id      | INTEGER       | FK courses.id               |
|                         | pattern        | VARCHAR(80)   | Curriculum pattern label    |
| syllabus_documents      | id             | INTEGER       | PK                          |
|                         | curriculum_id  | INTEGER       | FK curricula.id             |
|                         | year           | INTEGER       | Programme year              |
|                         | source_url     | VARCHAR(1000) | Reference document URL      |
|                         | status         | VARCHAR(30)   | Workflow / attendance state |
| curriculum_subjects     | id             | INTEGER       | PK                          |
|                         | curriculum_id  | INTEGER       | FK curricula.id             |
|                         | semester       | INTEGER       | Current / assigned semester |
|                         | code           | VARCHAR(60)   | Programme / subject code    |
|                         | credits        | NUMERIC(4, 1) | Published credit value      |
|                         | is_verified    | BOOLEAN       | Catalogue verification flag |
| campus_activities       | id             | INTEGER       | PK                          |
|                         | title          | VARCHAR(180)  | Activity title              |
|                         | coordinator_id | INTEGER       | FK users.id                 |
| activity_participations | id             | INTEGER       | PK                          |
|                         | activity_id    | INTEGER       | FK campus_activities.id     |
|                         | student_id     | INTEGER       | FK students.id              |
|                         | status         | VARCHAR(30)   | Workflow / attendance state |
|                         | reviewed_by    | INTEGER       | FK users.id                 |

Composite uniqueness: curriculum (course\_id, pattern); syllabus document (curriculum\_id, year); catalogue subject (curriculum\_id, semester, code); participation (activity\_id, student\_id). Catalogue verification and activity review remain explicit recorded states.

## 8 Academic Processing and Reports

### Subject allocation and attendance

Office creates teaching subjects for a course, curriculum and semester, then assigns the responsible faculty member. Staff receives the eligible cohort for that allocation. Attendance stores Present, Absent or Late per student and class date. The current report counts Present and Late as attended sessions.

Attendance percentage =  $\text{attended records} \div \text{recorded sessions} \times 100$ . With no sessions, progress remains pending. The implemented 75 percent warning is an application threshold that requires college confirmation before formal use.

### Marks and assignments

Internal and external marks must be non-negative and within configured maxima. NEP entry also requires a verified assessment link and matching component limits. The total is internal plus external marks; official NEP grade, SGPA and CGPA conversion is pending. Assignments store deadlines and maximum marks; submissions hold work, status and faculty feedback.

| Report               | Information produced                                                   |
|----------------------|------------------------------------------------------------------------|
| Students and Faculty | Identity, programme or department, contact and active status.          |
| Attendance           | Student, subject, date, status and recording faculty.                  |
| Marks                | Configured scores, totals and existing generic threshold-based result. |
| Assignments          | Assignment, subject, due date, maximum marks and submission count.     |
| Notifications        | Title, audience, creator, active status and timing.                    |

CSV and print-friendly views support review of stored records. The generic pass/fail export uses the subject passing threshold; it is not an official university result calculation.

## 9 Student and Staff Information Collection

Separate English Google Forms collect Student and Staff profiles into private response spreadsheets. As of 5 October 2026, nine student responses represent eight distinct BCA profiles after duplicate review. The local portal contains 19 students: eight submitted profiles and 11 generated identities, including five newly added Home Science students and FY/SY BCA students. Eleven staff profiles have teaching allocations.

| Form    | Required information                                                                                    | Optional information                       |
|---------|---------------------------------------------------------------------------------------------------------|--------------------------------------------|
| Student | Full name, email, college Enrollment/PRN, programme, class, semester, admission year and confirmation.  | Mobile, date of birth, gender and address. |
| Staff   | Full name, email, Employee ID, department, teaching subjects with programme/semester, and confirmation. | Qualification, mobile and joining date.    |

### Verified entry into PRAVA

- Collect responses using the relevant link. Passwords are not collected in Google Forms.
- Office checks identifiers, email uniqueness, programme, semester and admission year against official records.
- Prepare the exact Student or Staff UTF-8 CSV template, assigning a unique username to each verified person.
- Validate the import preview, then save the records. Imported accounts remain inactive until individual passwords and activation are completed by Office.
- Enter optional profile fields and teaching allocations separately. Student website sign-up remains a separate approval workflow.

Programme codes: BCA, BSC-FSN and BSC-TEXTILE; pattern: 2024 NEP. The local dataset includes 135 teaching subjects. Generated attendance, marks and coursework are linked only to presentation identities. Submitted profile values are preserved as authorised; unresolved details remain subject to college verification.

## 10 Input and Output Screens

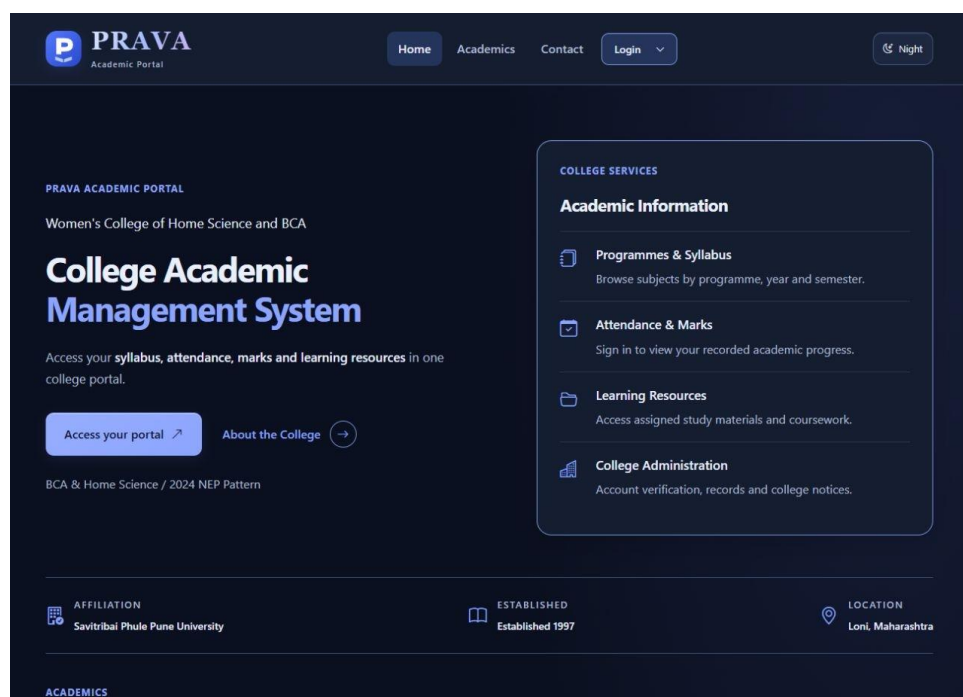


Figure 11 Current Home page in default Night mode

The header presents Home, Academics, Contact and Login. The revised heading identifies the College Academic Management System. Day mode is available from the theme control. The homepage presents academic services, programme information and college notices with functional headings.

Screenshots were refreshed on 5 October 2026 after the readability update. Named accounts display academic workflows. Generated academic entries illustrate local operation and are not official college results; the eight submitted BCA profiles retain their separate intake provenance.

# Login Dropdown and Programme Selection

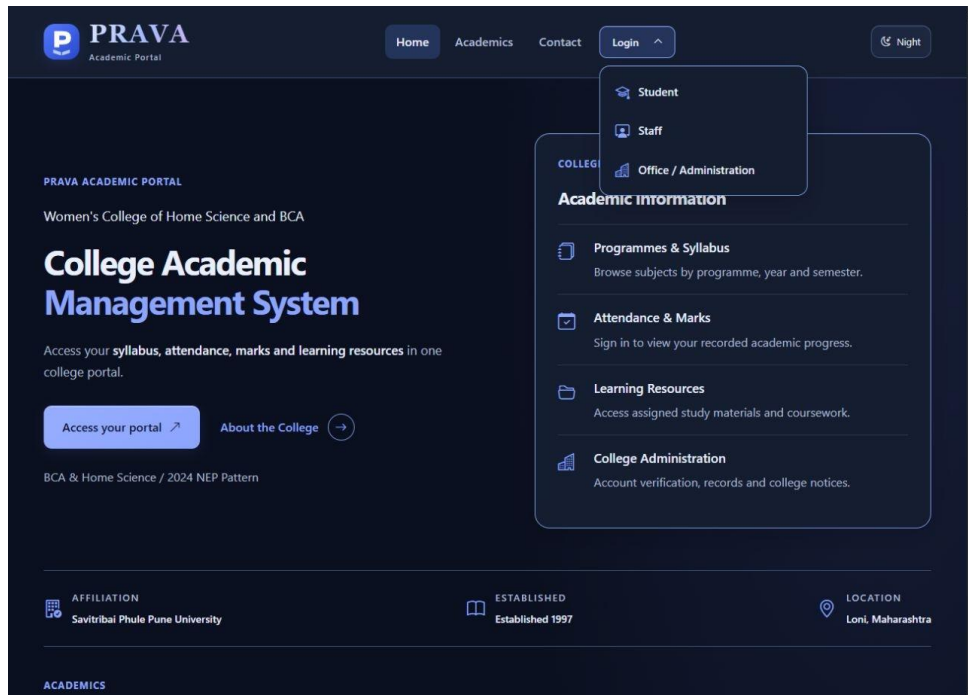


Figure 12 Login dropdown with Student Staff and Office options

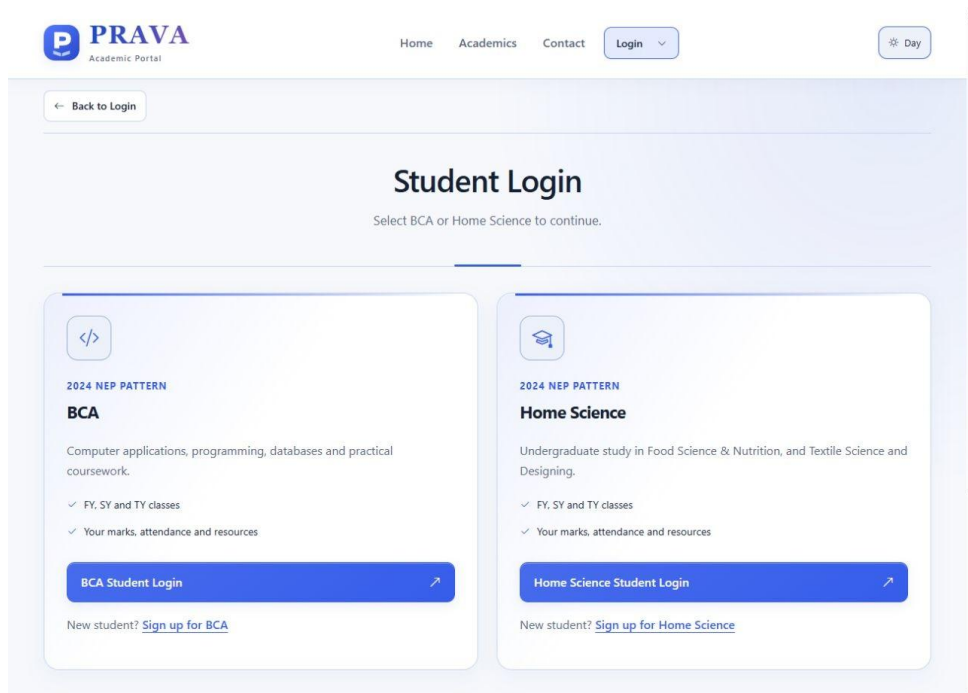


Figure 13 BCA and Home Science programme selection

Login opens a dropdown directly below the header control. Student continues to programme selection; the programme pages retain student sign-up links.

# Student Login and Website Sign Up

The screenshot shows the PRAVA Academic Portal interface. At the top, there is a navigation bar with the PRAVA logo, links for Home, Academics, Contact, and a Login button. A 'Day' toggle is also present. Below the navigation bar, a 'Back to programmes' link is visible. The main content area is split into two panels. The left panel, titled 'PRAVA ACADEMIC PORTAL', features a dark blue background with a graduation cap icon and the text 'BCA Student Login'. It instructs users to 'View your attendance, marks, subjects and assignments.' The right panel, titled 'BCA STUDENT PORTAL', contains the 'Sign In' form. It prompts users to 'Enter your college username or email and password.' and includes input fields for 'Username or Email' and 'Password'. There is a 'Remember Me' checkbox and a 'Need help?' link. A blue 'Sign in as BCA Student' button is at the bottom of the form. Below the button, a link for 'New to PRAVA? Create your student account' is provided. A note at the bottom states: 'Access is linked to your college account, programme and role.'

Figure 14 BCA student login form

The screenshot displays the 'Student Registration' form. At the top, the title 'Student Registration' is followed by a note: 'Join as a BCA or Home Science student. The college office checks your enrollment before activating your account.' The form is divided into two main sections. The left section, titled 'Registration Process', outlines three steps: 1. 'Your details' (Use your own email and college enrollment / PRN), 2. 'Office verification' (The office checks your programme, semester and enrollment), and 3. 'Account access' (Once approved, sign in with your chosen username and password). It also mentions 'Staff accounts are issued by the Administration office.' The right section contains the registration form itself, with fields for 'Full name', 'Choose a username', 'Your email', 'College enrollment / PRN', 'Your programme' (a dropdown menu showing 'Bachelor of Computer Applications'), 'Current semester' (a dropdown menu showing 'Semester 1'), 'Admission year' (a text field with '2026'), and 'Create password'. A 'Confirm password' field is also present. A blue 'Submit student sign-up' button is at the bottom. A small note at the bottom of the form states: '2024 NEP / Your password must contain at least 8 characters. Sign-up details are stored securely for Office review.'

Figure 15 Student sign-up form awaiting Office verification

The role and programme are explicit on login. Website sign-up stores a pending request; successful submission alone does not grant active account access.

# BCA Student Dashboard and Syllabus

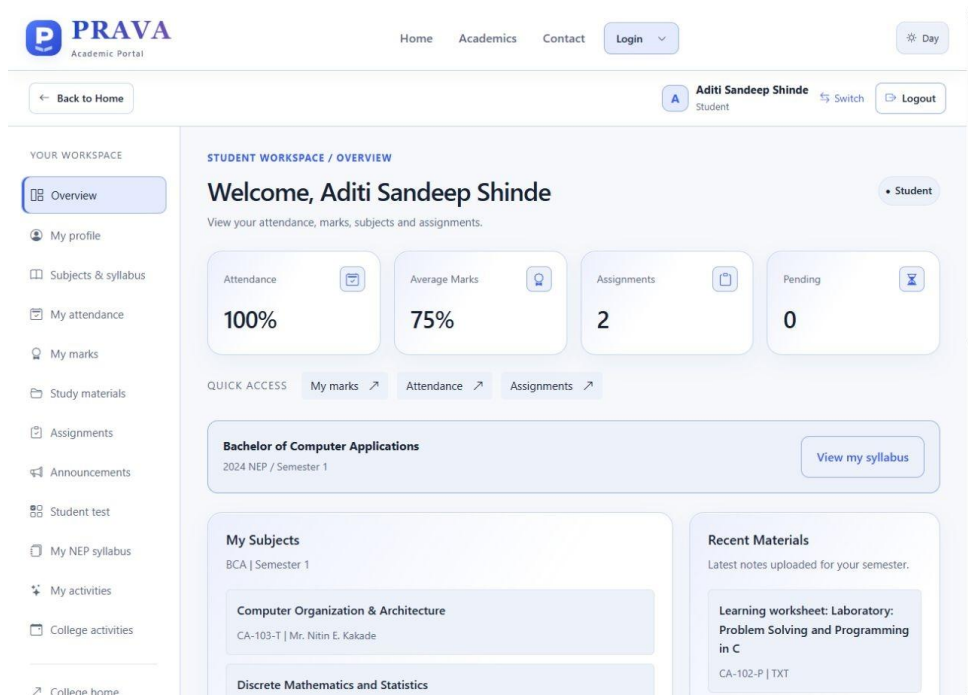


Figure 16 BCA student dashboard

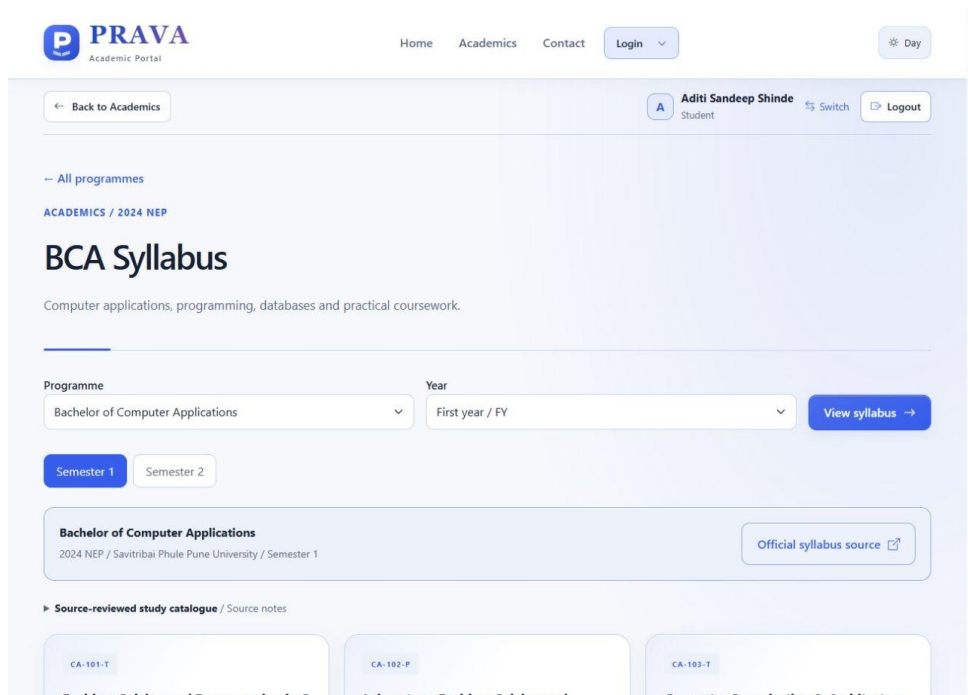


Figure 17 BCA NEP catalogue and official source link

Aditi Sandeep Shinde is the configured BCA presentation account. Its dashboard shows attendance, marks, coursework and allocated teaching subjects. The syllabus screen retains the source-linked NEP reference catalogue.

# Home Science Dashboard and Syllabus

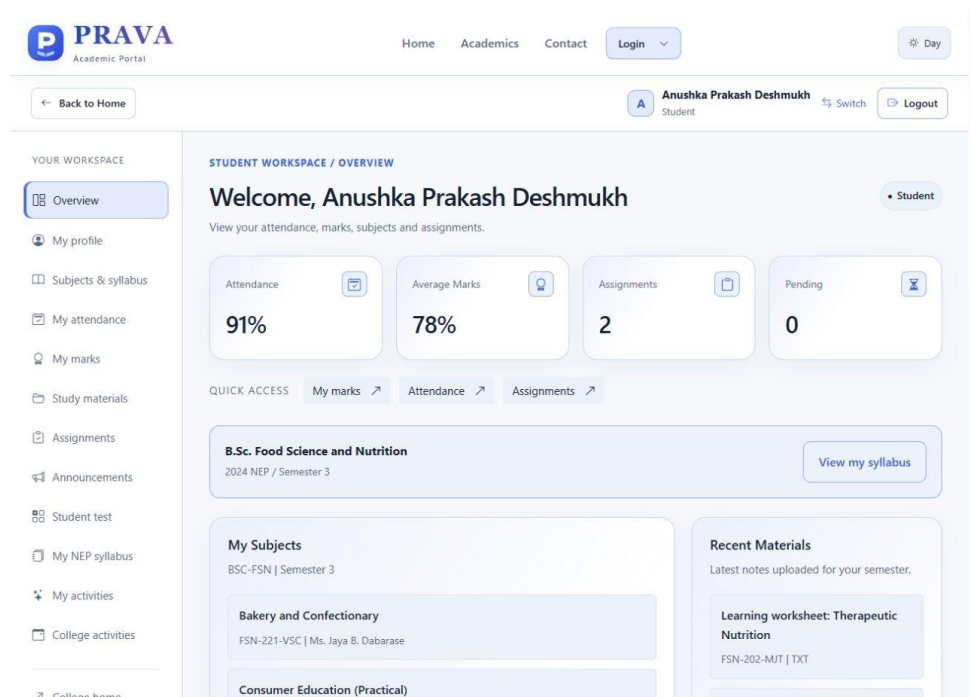


Figure 18 Home Science student dashboard

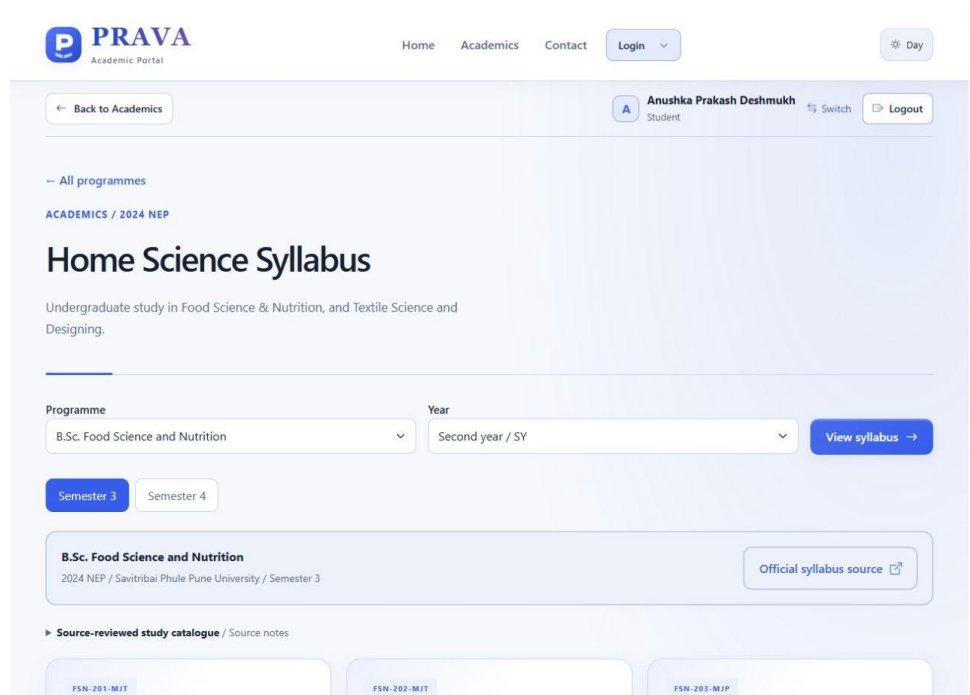


Figure 19 Home Science programme and semester catalogue

Anushka Prakash Deshmukh is configured for second-year Food Science and Nutrition. Home Science records also cover Textile Science and Designing, with FY, SY and TY students linked to their respective programmes.

# Staff Dashboard and Attendance

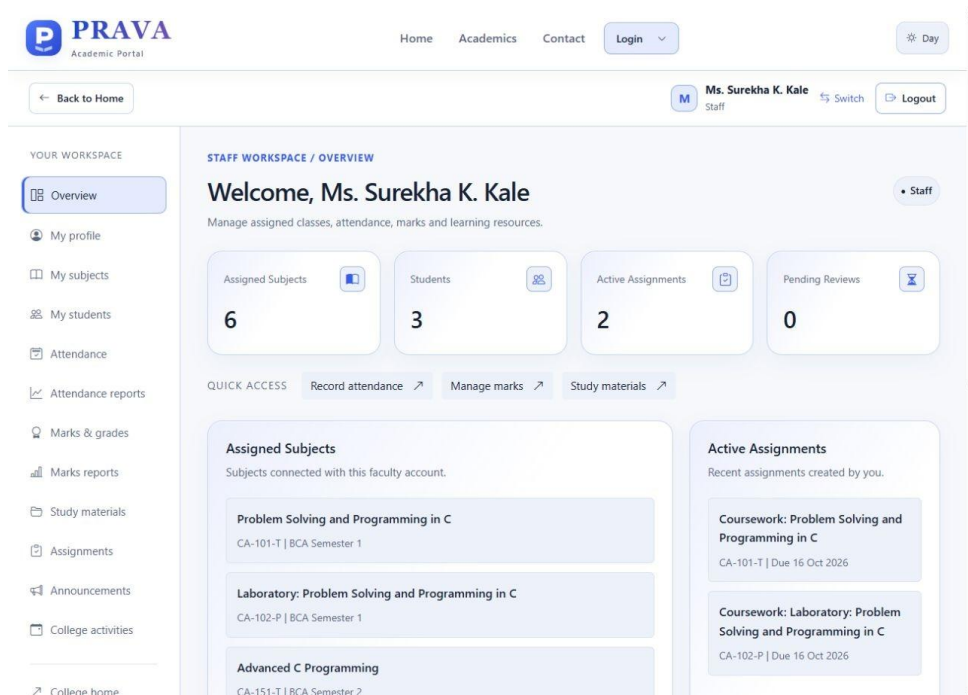


Figure 20 Staff dashboard with teaching summary

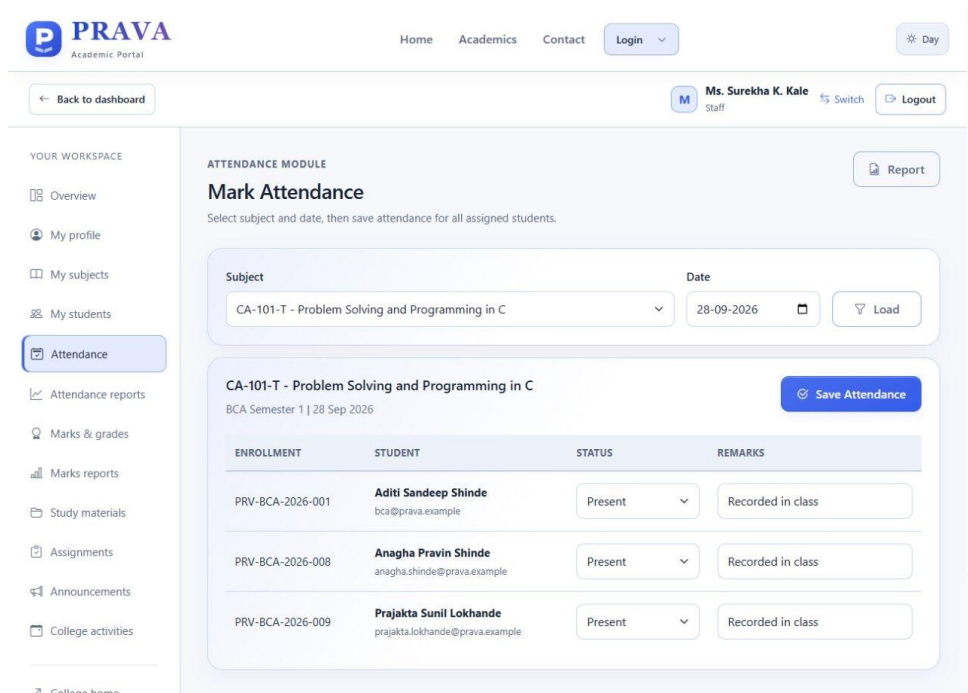


Figure 21 Attendance for an allocated teaching subject

Ms. Surekha K. Kale has six teaching allocations. The attendance screen loads the assigned BCA cohort and the selected class date. Academic entries and teaching allocations shown here are configured for the local project presentation.

# Office Dashboard and Registration Review

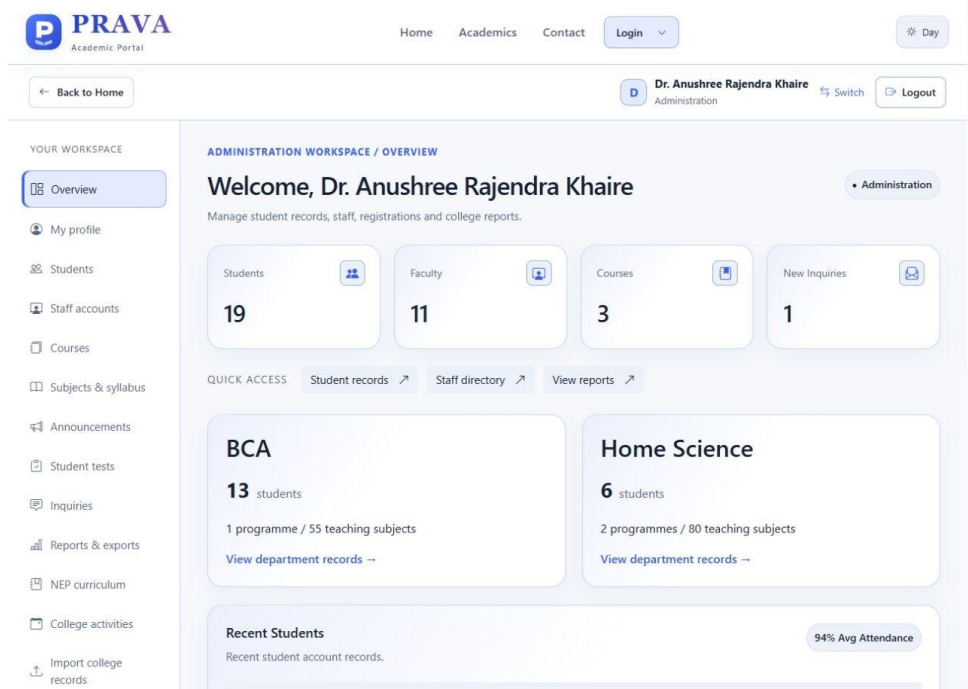


Figure 22 Office overview with BCA and Home Science departments

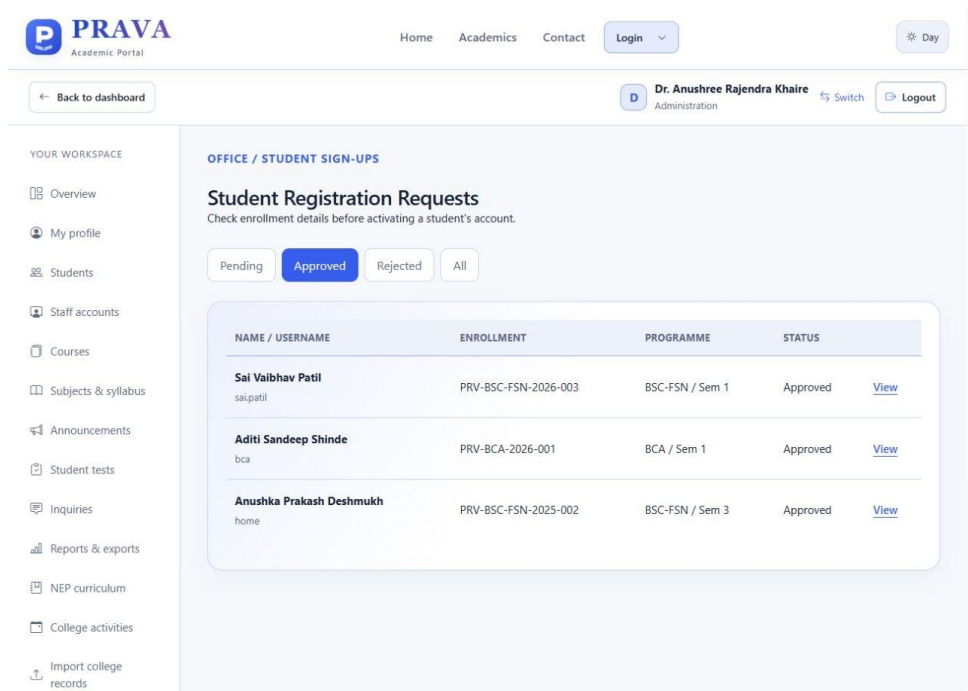


Figure 23 Office review queue for student sign-ups

The administration account displays Dr. Anushree Rajendra Khaire. The overview contains 19 student and 11 staff profiles, and the approved queue shows completed local registration records. Principal oversight also has a separate read-only account.

# Roster Import and Report Outputs

The screenshot shows the PRAVA Academic Portal interface. The top navigation bar includes 'Home', 'Academics', 'Contact', and a 'Login' button. The user is logged in as 'Dr. Anushree Rajendra Khaire' with an 'Administration' role. The left sidebar lists 'YOUR WORKSPACE' options: Overview, My profile, Students, Staff accounts, Courses, Subjects & syllabus, Announcements, Student tests, Inquiries, Reports & exports, NEP curriculum, College activities, and 'Import college records' (highlighted). The main content area is titled 'OFFICE / RECORDS' and 'Import College Records'. It includes instructions to 'Validate the complete file, review the preview, then save the records together.' Below this, there's a section 'Start with a template' with instructions to use official enrollment numbers, employee IDs, and email addresses. It provides two buttons: 'Student CSV template' and 'Staff CSV template'. A note states: 'New records stay inactive until the office sets an individual password and activates each account. Existing records are protected from duplicate imports.' There's a 'Record type' dropdown menu set to 'Students'. Below that, a 'College roster (UTF-8 CSV, up to 200 rows)' section has a 'Choose File' button and a 'No file chosen' status. A 'Validate & preview' button is at the bottom.

Figure 24 Separate Student and Staff CSV import templates

The screenshot shows the PRAVA Academic Portal interface. The top navigation bar is the same as Figure 24. The left sidebar highlights 'Reports & exports' under 'YOUR WORKSPACE'. The main content area is titled 'REPORTS MODULE' and 'Reports'. It includes a 'Print' button and instructions to 'Generate CSV reports for academic and master data.' Below this, there are six report cards, each with a 'Download CSV' button: 'Student Report' (Enrollment, contact, course, semester, and active status), 'Faculty Report' (Employee ID, department, contact, subject count, and active status), 'Attendance Report' (Daily attendance rows with student, subject, status, and faculty), 'Marks Report' (Subject marks, totals, grades, and pass/fail result), 'Assignment Report' (Assignments, due dates, maximum marks, and submission counts), and 'Notification Report' (Announcements, target filters, creator, and active status).

Figure 25 Administrative report and CSV export options

The import workflow validates a complete file before commit. Reports and exports use the stored records and applicable access scope.

# Student Marks and Coursework

**PRAVA**  
Academic Portal

Home Academics Contact Login

Aditi Sandeep Shinde  
Student

Back to dashboard

YOUR WORKSPACE

- Overview
- My profile
- Subjects & syllabus
- My attendance
- My marks**
- Study materials
- Assignments
- Announcements
- Student test
- My NEP syllabus
- My activities
- College activities
- College home

**STUDENT WORKSPACE / RESULTS**

**My marks**

Your academic results, one subject at a time.

Aditi Sandeep Shinde  
PRV-BCA-2026-001 / BCA / Semester 1

7 results

**Subject-wise results**

Internal and external marks, totals, and grades recorded by your teaching staff.

| SUBJECT                                                                 | EXAM TYPE     | INTERNAL | EXTERNAL | TOTAL   | GRADE   |
|-------------------------------------------------------------------------|---------------|----------|----------|---------|---------|
| <b>Computer Organization &amp; Architecture</b><br>CA-103-T             | Semester Exam | 11       | 25       | 36 / 50 | Pending |
| <b>Discrete Mathematics and Statistics</b><br>CA-105-T                  | Semester Exam | 14       | 25       | 39 / 50 | Pending |
| <b>HTML and Web Page Designing</b><br>SEC-101-CA                        | Semester Exam | 11       | 25       | 36 / 50 | Pending |
| <b>Laboratory: Computer Organization &amp; Architecture</b><br>CA-104-P | Semester Exam | 15       | 25       | 40 / 50 | Pending |

Figure 26 Student subject results with component marks

**PRAVA**  
Academic Portal

Home Academics Contact Login

Aditi Sandeep Shinde  
Student

Back to dashboard

YOUR WORKSPACE

- Overview
- My profile
- Subjects & syllabus
- My attendance
- My marks
- Study materials
- Assignments**
- Announcements
- Student test
- My NEP syllabus
- My activities
- College activities
- College home

**STUDENT MODULE**

**My Assignments**

View assignment details, submit work, and check grades.

| TITLE                                                        | SUBJECT  | DUE DATE    | MAX MARKS | STATUS | MARKS | FEEDBACK                                                            | ACTION   |
|--------------------------------------------------------------|----------|-------------|-----------|--------|-------|---------------------------------------------------------------------|----------|
| Coursework: Problem Solving and Programming in C             | CA-101-T | 16 Oct 2026 | 20        | Graded | 16    | Well organised work. Add clearer references in the next submission. | Download |
| Coursework: Laboratory: Problem Solving and Programming in C | CA-102-P | 16 Oct 2026 | 20        | Graded | 16    | Well organised work. Add clearer references in the next submission. | Download |

Figure 27 Submitted coursework with marks and faculty feedback

The student results page shows stored subject components and totals. Official NEP grade calculations remain pending. Coursework lists due dates, submission status, awarded marks and feedback for the signed-in student.

# Staff Review and Grading

**PRAVA** Academic Portal

Home Academics Contact Login

Ms. Surekha K. Kale Staff Logout

← Back to assignments

YOUR WORKSPACE

- Overview
- My profile
- My subjects
- My students
- Attendance
- Attendance reports
- Marks & grades
- Marks reports
- Study materials
- Assignments**
- Announcements
- College activities
- College home

**ASSIGNMENT MODULE**

Coursework: Problem Solving and Programming in C

CA-101-T / Due 16 Oct 2026, 05:00 PM

| STUDENT                 | ENROLLMENT       | STATUS | SUBMITTED             | MARKS | FEEDBACK                                                            | ACTION                                        |
|-------------------------|------------------|--------|-----------------------|-------|---------------------------------------------------------------------|-----------------------------------------------|
| Aditi Sandeep Shinde    | PRV-BCA-2026-001 | Graded | 03 Oct 2026, 10:00 AM | 16    | Well organised work. Add clearer references in the next submission. | <a href="#">Download</a> <a href="#">Edit</a> |
| Anagha Pravin Shinde    | PRV-BCA-2026-008 | Graded | 03 Oct 2026, 10:00 AM | 16    | Well organised work. Add clearer references in the next submission. | <a href="#">Download</a> <a href="#">Edit</a> |
| Prajakta Sunil Lokhande | PRV-BCA-2026-009 | Graded | 03 Oct 2026, 10:00 AM | 17    | Well organised work. Add clearer references in the next submission. | <a href="#">Download</a> <a href="#">Edit</a> |

Figure 28 Faculty review of submissions for an allocated subject

**PRAVA** Academic Portal

Home Academics Contact Login

Ms. Surekha K. Kale Staff Logout

← Back to dashboard

YOUR WORKSPACE

- Overview
- My profile
- My subjects
- My students
- Attendance
- Attendance reports
- Marks & grades
- Marks reports
- Study materials
- Assignments**
- Announcements
- College activities
- College home

**ASSIGNMENT MODULE**

Grade Submission

Aditi Sandeep Shinde / Coursework: Problem Solving and Programming in C

**Marks Obtained**

16

Maximum marks: 20

**Feedback**

Well organised work. Add clearer references in the next submission.

Cancel Save Grade

Figure 29 Assignment score and feedback entry

The assigned faculty member reviews the submitted file, awards a score within the assignment maximum, and records feedback. The displayed 16 out of 20 and review text are existing local presentation records.

## 11 Testing and Verification

All 63 automated tests passed on 5 October 2026 in 26.480 seconds using isolated fixtures. The run used repository baseline 4abb237 with local typography improvements. The controlled error-route test intentionally logs an exception while checking the friendly error response.

| Test module     | Tests | Result |
|-----------------|-------|--------|
| college_upgrade | 15    | Passed |
| onboarding      | 11    | Passed |
| phase14         | 12    | Passed |
| public_homepage | 20    | Passed |
| student_test    | 5     | Passed |
| Total           | 63    | Passed |

| Representative case            | Expected and verified result                                                                  |
|--------------------------------|-----------------------------------------------------------------------------------------------|
| Login and switching            | Role/programme checks; matching dashboard; failed switch preserves prior identity.            |
| Registration and imports       | Pending review; duplicate conflicts rejected; transaction rollback prevents partial accounts. |
| Permissions and security       | Principal write restrictions; Staff subject ownership; CSRF and inactive account checks.      |
| Academics and communication    | NEP score bounds; curriculum publication status; notice audiences; activity ownership.        |
| Public pages and student tests | Separate pages, valid contact storage, protected student tests and correct score handling.    |

Browser checks on 5 October 2026 confirmed all four main login flows and refreshed the report screenshots. Day and Night layouts were inspected at desktop and mobile widths. Key labels, form fields and table text use larger type; wide tables remain horizontally scrollable within their panels.

## Input Validation Examples

The following examples explain implemented safeguards. They were checked through the isolated automated test suite on 5 October 2026; no invalid records were added to the working college database.

### Login and programme checks

An incorrect password returns “Invalid username/email or password.” A BCA account cannot sign in through the Home Science portal. Inactive accounts are blocked, and an unsuccessful account switch preserves the previous valid identity.

### Duplicate registration and import

Submitting the same student sign-up twice leaves one request. An existing enrollment / employee number is rejected by the CSV import preview. Approval and import commit repeat uniqueness checks, so a conflicting record prevents partial account creation.

### Marks and assignment bounds

| Validation example                       | Result                                     |
|------------------------------------------|--------------------------------------------|
| Internal maximum 15; enter 16            | Rejected: component exceeds its limit.     |
| External maximum 35; enter 36            | Rejected: component exceeds its limit.     |
| Assignment maximum 100; enter -1 or 101  | Rejected: negative or above-maximum score. |
| Student from a different academic cohort | Rejected: no eligible subject membership.  |

### Verification evidence

test\_phase14 covers login and assignment bounds; test\_onboarding covers repeated sign-up and approval conflicts; test\_college\_upgrade covers NEP component limits, roster duplicates and curriculum ownership. All 63 tests passed in 26.480 seconds.

## 12 Security and Data Maintenance

### Access and consistency

Passwords are stored as hashes. Protected routes check authentication, active status, roles and academic ownership. State-changing requests require CSRF tokens, including Logout. Imports and registration approval revalidate duplicates before commit. Unique keys and foreign keys complement service-level checks; selected actions are recorded in `activity_logs`.

### Persistent storage and recovery

| Location               | Stored information                                                      |
|------------------------|-------------------------------------------------------------------------|
| instance/prava.sqlite3 | Users, profiles, academic records, inquiries, review and activity data. |
| instance/.local-secret | Persistent local session secret.                                        |
| instance/backups/      | Backups created by the launcher before detected schema upgrades.        |
| app/static/uploads/    | Profile images, materials, assignments and submitted work.              |
| Google response Sheets | Student and Staff intake responses; separate from the PRAVA database.   |

A complete backup preserves the database and required uploaded files. Stop the local server for a file-copy backup or use a consistent database backup procedure. Reopening the same installation retains its records. A newly extracted ZIP creates a separate installation, so populated folders should be preserved during updates.

### Public deployment

The launcher binds to 127.0.0.1. Public hosting requires production settings, HTTPS, persistent database and file storage, private secrets, restricted administration and tested recovery. Local presentation credentials must be disabled for real operation. Repository publication delivers source code; it does not publish the local database or deploy a public service.

## **13 Advantages Limitations and Future Work**

### **Advantages**

- Public information and academic work are available through one consistent portal.
- Programme-aware student access and explicit Staff allocations keep records in the correct academic context.
- Office review, duplicate checks and transactional imports support controlled account creation.
- Source links, review labels, persistent local storage and CSV exports support demonstration and record review.

### **Current limitations**

The catalogue has 131 entries: 123 source-reviewed public entries and eight drafts. Shared Home Science content appears under both specialisations. The BCA-linked source is titled B.Sc. Computer Applications; SY is marked draft. Twelve local Home Science TY practice topics support presentation without being labelled verified university syllabus. Detailed TY sources and some assessment details await confirmation.

Official NEP grade, SGPA/CGPA and individual elective enrolment are not implemented. The project does not automate email after Office approval. The Google Forms require review and manual conversion to import templates. The small local presentation dataset does not establish institutional capacity or full-semester operation.

### **Future enhancements**

Priorities are approved batch-specific syllabus updates, elective registration, institution-validated grade calculations, controlled account recovery and registration notifications. Hosted operation should add verified backup restore, accessibility review and performance testing with representative data.

## 14 Project Cost and Conclusion

### Illustrative cost estimate

The following is an academic planning example in INR, not recorded expenditure or logged effort. It assumes an existing laptop and connectivity. Team-hours are the combined effort of both members and are not multiplied by two again.

| Planning item               | Calculation                                 | Value INR |
|-----------------------------|---------------------------------------------|-----------|
| Requirements and design     | 40 hours $\times$ 100                       | 4,000     |
| Implementation              | 70 hours $\times$ 100                       | 7,000     |
| Verification                | 26 hours $\times$ 100                       | 2,600     |
| Documentation               | 24 hours $\times$ 100                       | 2,400     |
| Total effort value          | 160 hours $\times$ 100                      | 16,000    |
| Printing budget             | Up to 45 pages $\times$ 3 $\times$ 2 copies | 270       |
| Binding budget              | 150 $\times$ 2 copies                       | 300       |
| Local hosting fee           | No paid hosting for local presentation      | 0         |
| Illustrative combined value | 16,000 + 270 + 300                          | 16,570    |

### Conclusion

PRAVA demonstrates a working college portal with clear role entry, programme-aware student access, Office-reviewed onboarding and academic tools. The relational design separates identity, teaching allocation, curriculum references and academic records. The current interface, 22-table schema, portable launcher and 63 passing tests support the implementation described in this report.

The project provides practical experience in web development, relational modelling, validation and software verification. Further work should complete approved academic rules and source review, then evaluate real operational data and hosting requirements.

## 15 Bibliography

Project implementation and technical references. The software versions are the repository requirements; curriculum references are preserved from the application source review.

### 1. PRAVA source repository

kvaishu0703/Prava\_Edu. Baseline 4abb237 with local typography improvements; dataset, interface and 63 automated tests verified on 5 October 2026. README, models, services, templates and tests.  
[https://github.com/kvaishu0703/Prava\\_Edu](https://github.com/kvaishu0703/Prava_Edu)

### 2. Python documentation

Python Software Foundation. Python 3.12 documentation.  
<https://docs.python.org/3.12/>

### 3. Flask documentation

Pallets Projects. Application factories, configuration, blueprints and testing. Project dependency Flask 3.0.3.  
<https://flask.palletsprojects.com/en/stable/>

### 4. SQLAlchemy ORM documentation

SQLAlchemy. Object relational mapping and model relationships.  
<https://docs.sqlalchemy.org/en/20/orm/>

### 5. College information

Women's College of Home Science and BCA, Loni. Institutional information and academic references recorded by the project.  
<https://www.pravarahomesciencebca.in/>

### 6. Google Forms setup

Google Apps Script Form service. Separate student/staff forms, validation, publishing and spreadsheet destinations; created 30 September 2026 and updated to English on 4 October 2026.  
<https://developers.google.com/apps-script/reference/forms/form>

## Curriculum Source References

**7. SPPU FY B.Sc. Computer Applications syllabus linked by the college BCA department.**

[http://www.pravarahomesciencebca.in/documents/STUDENT%20CORNER/Syllabus/BCA/B.%20Sc.%20%28Computer%20Applications%29\\_14062024.pdf](http://www.pravarahomesciencebca.in/documents/STUDENT%20CORNER/Syllabus/BCA/B.%20Sc.%20%28Computer%20Applications%29_14062024.pdf)

**8. SPPU SY B.Sc. Computer Applications draft hosted by Pratibha College.**

[https://pcccs.org.in/wp-content/uploads/2026/03/SY-B.Sc\\_.CA\\_.pdf](https://pcccs.org.in/wp-content/uploads/2026/03/SY-B.Sc_.CA_.pdf)

**9. SPPU FY B.Sc. Home Science syllabus linked by the college.**

[http://www.pravarahomesciencebca.in/documents/STUDENT%20CORNER/Syllabus/Home%20Science/syllabus/B.Sc.%20%28Home%20Science%29\\_07062024.pdf](http://www.pravarahomesciencebca.in/documents/STUDENT%20CORNER/Syllabus/Home%20Science/syllabus/B.Sc.%20%28Home%20Science%29_07062024.pdf)

**10. SPPU SY B.Sc. Home Science syllabus on the university circulars site.**

[http://collegecirculars.unipune.ac.in/sites/documents/Syllabus2025/S.Y.B.Sc.%20\(Home%20Science\)\\_25082025.pdf](http://collegecirculars.unipune.ac.in/sites/documents/Syllabus2025/S.Y.B.Sc.%20(Home%20Science)_25082025.pdf)

The catalogue records its source-check date as 20 September 2026. Source title, pattern, effective year and review notes must be interpreted together. A college department label may differ from a university document title. Formal academic use requires confirmation that each document applies to the student batch.

The report preserves known source qualifications and current implementation status. It does not convert draft or incomplete syllabus content into a confirmed university scheme.

## Appendix Local Installation and Demonstration

- Open [https://github.com/kvaishu0703/Prava\\_Edu](https://github.com/kvaishu0703/Prava_Edu) and choose Code → Download ZIP, then Extract All.
- Preserve any existing populated project folder. A new folder has its own local database.
- Install supported 64-bit Python; the README recommends 3.12 with Add Python to PATH.
- Open the inner folder containing START-PRAVA.cmd and double-click it. Internet is needed for first dependency setup.
- Keep the server window open. The browser opens <http://127.0.0.1:5000/>. Stop the server with Ctrl+C.

PowerShell: right-click the project folder → Open in Terminal. Run `.\START-PRAVA.cmd`. If port 5000 is busy, run `.\START-PRAVA.cmd -Port 5001` and open <http://127.0.0.1:5001/>. The `.\` prefix selects the file in the current folder.

| Role                    | Login ID | Password  |
|-------------------------|----------|-----------|
| BCA student             | bca      | bca123    |
| Home Science student    | home     | home123   |
| Staff                   | staff    | staff123  |
| Office / Administration | office   | office123 |

These credentials are for local presentation only. Verified college accounts use individual credentials. The local accounts are blocked when `PRAVA_DEMO_MODE` is disabled or production configuration is active.

Presentation order: Home and theme → Login dropdown → BCA and Home Science dashboards → Staff academic tools → Office review, import and reports. Use Logout or Switch account between roles. README.md contains the setup and maintenance instructions.